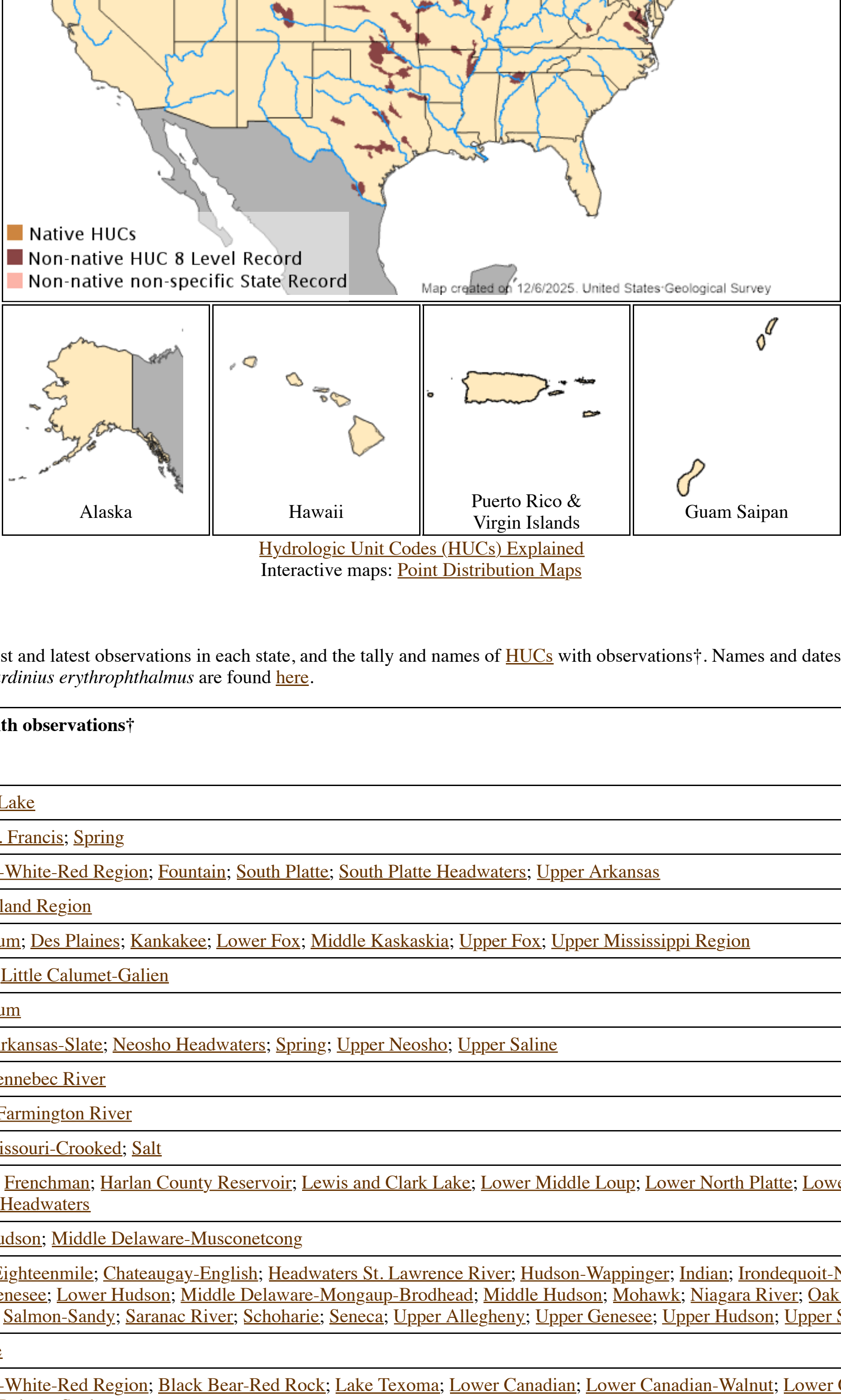






PA	1991	2012
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SD	1992	2023	12	Upper: Snake; Upper Minnesota; Vermilion
TX	1989	2001	12	International Falcon; Reservoir: Lake Texoma; Lower Brazos; Lower Colorado-Cummins; Lower Sulphur; Middle Brazos-Lake Whitney; North Concho; San Marcos; Upper Guadalupe; Upper San Antonio; Upper Trinity; West Fork San Jacinto
Vt	1901	2025	κ	Black River; Connecticut River; Lake Champlain; Merrimack River; Richelieu; Missouci; Dover

WV	1989	1993	4	Lower Potomac: Lynnhaven-Potomac; Middle Potomac: Anacostia-Ocquan; Pamunkey
WV	1991	1991	1	Lower New
WI	1916	2000	5	Door Keweenaw; Lake Michigan; Lake Winnebago; Middle Rock; Upper Rock

Table last updated 12/8/2025

* Populations may not be currently present.

Ecology: Rudd is a benthic cyprinid fish that prefers water up to 3 meters deep and temperatures between 2 and 22°C (Gerstmeier and Romig 1998; Glankins-Lewczuk et al. 2016; Juza et al. 2018). Abundance decreases with depth, and Rudd populations benefit from artificial and natural coastal refuges that help them avoid predation (Lemmens et al. 2016; Vorkoback and Stankovic 2016). Rudd were previously described as a littoral species that prefers benthic habitats (Johansson 1987; Lake et al. 2002); however, recent studies of Rudd populations in the Niagara River found higher abundance, growth, and reproductive success in lotic versus lentic habitats (Kapusinski et al. 2012a).

Rudd has omnivorous feeding habits that can change with temperature and season. Generally, their diet includes macrophytes, bryophytes, and filamentous algae along with fish and some other animal material and detritus (Nurminen et al. 2003; Guinan et al. 2015). Rudd in Lake Banyoles, an oligotrophic, karstic lake that was more zooplanktivorous in spring and autumn and less in summer and showed a size-dependent diet shift from macrozooplankton to macroinvertebrates at larger sizes (García-Berthou and Moreno-Amich 2007; Guinan et al. 2015) showed Rudd in the upper Niagara River exhibited a diet switch from piscivory in early spring to zooplanktivory in summer, but this switch was related to increased macrophyte abundance and warmer water temperatures. The shift from piscivory to herbivory typically occurs once temperatures exceeded ~16°C, allowing for food availability to avoid competition with other fishes or to avoid predation by piscivorous fishes (Johansson 1987; Eklov and Hamrin 1989; Kapusinski et al. 2012a). Another reason for a shift in diet at higher temperatures is that cellulase efficacy was reduced at temperatures greater than 15°C (Vejrkova et al. 2016). Feeding intensity of Rudd was also positively correlated with water temperature (Kapusinski et al. 2012a)

Reproduction occurs from April to August, typically beginning once water temperatures exceed 16°C (Domagala et al. 2020). Adhesive eggs are laid among submerged vegetation in warmer, calmer, waters near shore. Spawning temperature ranges from 3,500–22,000 eggs. In the upper Niagara River, the presence of age-0 Rudd was greater in lotic sites and poorer in ones without measurable flow, suggesting reproduction can be successful in lotic habitats (Kapusinski et al. 2012a). Maturation occurs at 2 to 3 years when the fish is 90 to 150 mm in total length. Maximum life span can reach 17 years (Pfleger 1997).

Rudd are consumed by various predatory fish, including Pike (*Esox lucius*), Pike-Perch (*Sander lucoperca*), and Webs Catfish (*Silurus glanis*) (Gurbik et al. 2015; Djani et al. 2019). Rudd have similar energy density to other Great Lakes fishes and may be a useful food source (Forzono et al. 2017).

Means of Introduction: Bait bucket release seems to be the primary mechanism by which Rudd have gained access into open waters. However, the history of its introduction is complex and only partly known. The species apparently entered the United States during two widely separated periods of introduction. It was initially brought to this country either in the late 1800s or early 1900s. A second period of introduction presumably began in the late 1960s or early 1970s. The first verifiable U.S. record dates to 1916. In that year, some 300 specimens were obtained from the New York aquarium by B. O. Webster, the Wisconsin superintendent of fisheries, were transplanted to Lake Oconomowoc, Wisconsin (Cahn 1927; Greene 1935; Myers (1925) reported on a population in Central Park Lake in New York City. The Central Park population, now extirpated, may have originated from Copake Lake, Rockfall-Jensen Kill system, which also supported a population in 1916 and possibly earlier (M. N. Feinberg, personal communication to Burkhead). The origin of the very first New York fish is not known; however, because Rudd is popular in Europe as a food and game species, it may have been introduced for both purposes. The Central Park population also may have been the source for the transplant to a New Jersey park just across the Hudson (Myers 1925; Greeley 1935).

The early 1950s capture of Rudd by C. R. Robins in Cascadilla Creek, Tompkins County, New York, appears to be the only and last known record from the creek (Courtney et al. 1984; Smith 1985). The origin of this population is not documented, but the relatively close Rockfall-Jensen Kill population is a logical possibility. The earliest of the more recent records is a 1973 capture of Rudd from Cobosseecontee Lake in Maine. Neither year nor origin of the Maine introduction is known. At about the same time, a bait dealer in Suffolk, Virginia, began rearing Rudd for bait. The source of the Suffolk, Virginia fish farmer's stock is uncertain. The interest in bait introduction of Rudd dramatically intensified in the early 1980s. The central Arkansas region of Lonoke and Prairie counties, an area known for its active fish farming industry, apparently became the largest producer of Rudd in the United States. It appears that the greatest dispersal of Rudd has been through interstate traffic rather than direct European import. In fact, much of its recent culture and spread can be attributed to its popularity as a bait among Striped Bass *Morone saxatilis* anglers. As a result, Rudd have been widely introduced through a combination of bait bucket releases, escapes from aquaculture facilities and farm ponds, and, presumably, by dispersal from various points of introduction (e.g., Burkhead and Williams 1991). Although many Rudd introductions are considered accidental, it is likely that Rudd also have been intentionally released into public waters during the past few decades.

Status: This species has been recorded as introduced to 23 US states and the Canadian province of Ontario. In a number of other states it has been used as a bait fish, but there is as yet no record of it being found in open waters. Available data indicate established populations still survive in Maine and New York, and, more recently, evidence indicates it is established in Massachusetts, Nebraska, and South Dakota. The species had breeding populations in New Jersey, Wisconsin, and Kansas, but it is apparently no longer extant in these states. It has been established in Lake Cobosseecontee, Maine, since 1973 (Courtney et al. 1984; Burkhead and Williams 1991; Kirchels 1994). It has been established in the Rockfall-Jensen Kill drainage, New York, since the first half of the 1900s (Smith 1985). Probably in reference to that region, Schradi (1986) listed Rudd as present in the Hudson River drainage in the northern Appalachian region. Other New York populations are considered extirpated (Courtney et al. 1984; Smith 1985); however, Mills et al. (1993) stated that an established population was discovered in Onondica Lake, New York, in the Lake Ontario drainage, in 1990. It may also be established in the St. Lawrence River and Lake Ontario, but that conclusion awaits confirmation. Hartel et al. (1996) stated that the presence of both young and adults over several years indicated that it is reproducing in the lower Charles River in Massachusetts. According to Zafina (personal communication), Rudd is considered to be established in one or more lakes in Nebraska as of 1998. Presumably this includes Lake Oahe/Lake, there also is evidence of Rudd reproduction in parts of western South Dakota during the summer of 1997; several year classes were discovered in Pactola Reservoir, Sheraton Lake, and Nevall Lake (Unkenholz, personal communication). The species was established and later became extirpated in New Jersey and Wisconsin, however, it is still considered rare in Oconomowoc Lake, WI (Becker 1983; Courtney et al. 1984; Courtney and Williams 1992; Kapusinski et al. 2012b). There has been at least one established population in Kansas. In Spring 1996, a large breeding population was discovered in a 0.6-acre farm pond in the Deep Creek-Kansas River drainage near Manhattan, Riley County. However, that population was exterminated by Kansas Department of Wildlife and Parks personnel in Summer of 1996; as of early 1998, there has been no record of Rudd in sites downstream of the pond (Bever, personal communication). Although they mapped its distribution in Kansas, Cross and Collins (1995) did not provide information on reproductive status. In general, the past literature on Rudd distribution and status is fragmentary and somewhat contradictory. Courtney et al. (1986, 1991), Courtney and Stauffer (1990), and Courtney and Williams (1992) considered it established in the Maine and New York. Similarly, Page and Burr (1991) stated that it was established in Maine and in the lower Hudson River drainage in New York. In a relatively recent work, Courtney (1993) indicated that the Rudd was established in Maine, New York, Kansas, and presumably Nebraska (postal abbreviation given as NB). Jenkins and Burkhead (1994), citing unpublished information, stated that Rudd had become established in Indiana, New York, and Maine. However, we have found no evidence that the Rudd was ever introduced to Indiana waters. It is likely that the inclusion of Indiana was an error (Burkhead, personal communication). Muenke (1990) incorrectly cited unpublished information in stating that Rudd apparently had become established in Alabama. In many states, it has been recorded but not known to be reproducing. For instance, Jenkins and Burkhead (1994) stated that it is unknown whether escapees have established a population in Virginia. Although it has been reported from several reservoirs in Missouri, Pfeleger (1997) stated there is no evidence that Rudd is established in the state. Although we have a relatively recent report of this species from Connecticut (Jones, personal communication); Whitworth (1996) stated that the species has not been found in that state. Biologists in Arkansas say it is not established in that state (N. Stone and S. Barkley, pers. comm. 2007).

Great Lakes:
Colonizing, with populations reproducing and overwintering at self-sustaining levels in Lake Erie, Lake Ontario, and Lake Michigan.

Impact of Introduction: Largely unknown. In a laboratory setting, Burkhead and Williams (1991) demonstrated that Rudd readily hybridize with native golden shiner, *Notemigonus crysoleucas*, a primary forage species of many native game fishes. As such, the probability exists that Rudd introduced to open waters will hybridize with golden shiner, with unknown consequences to wild populations of the native species. First generation hybrids and their offspring should show heterosis (or hybrid vigor), but the "genetic pollution" in subsequent generations could prove detrimental due to a variety of factors, for instance spawning behavior and recruitment success, and general loss of fitness (Burkhead and Williams 1991; Courtney and Williams 1992). Nevertheless, the interactions of the two species in nature are not known (Burkhead, personal communication). Cadwallader (1977) reviewed the literature and concluded that Rudd is established in the state. Although we have a relatively recent report of this species from Connecticut (Jones, personal communication); Whitworth (1996) stated that the species has not been found in that state. Biologists in Arkansas say it is not established in that state (N. Stone and S. Barkley, pers. comm. 2007).

Remarks: According to Courtney and Williams (1992) and Gilbert (1988), Rudd first appeared in the United States during the late 1800s. Main support for their conclusion is based on a footnote of Hubbs (1921). In his footnote, reference is made to a brief description by Bean (1897) of a variety of golden shiner *Notemigonus crysoleucas* from Central Park Lake, New York City, which Hubbs concluded to be, in fact, introduced *Scardinus carolinianus* (Courtney and Williams 1991). The "permanent vermilion" or "pinkish red" of the pectorals, ventrals and anal" described by Bean for specimens apparently held in Central Park Lake, New York City, is repeated by Courtney and Williams (1992) using the same description as in the 1897 paper, elevated the race to subspecies (i.e., *Abramis chrysolaemus roseus*). The principal character on which the new form was based was the blood-red pigmentation of the fins (hence the name roseus). In his investigation on the history of Rudd in the United States, Burkhead (personal communication) was unable to locate types for the subspecies, but he did find a single *N. crysoleucas*, dated 1894, from Central Park Lake, which he examined and concluded as typical in every respect. According to Burkhead, the available information does not support or refute Hubbs' comments. Furthermore, since Hubbs' remarks presumably are based solely on Bean's color description, Burkhead concluded that it seems prudent to discard Hubbs' inference.

Courtney et al. (1986) described the distribution of Rudd in the eastern United States, listing it among the species with declining populations. However, Courtney et al. (1991) and Burkhead and Williams (1991) reviewed Rudd introductions and documented the more recent and rapid spread of the species as a result of its wide use as a bait fish for Striped Bass. Many states now outlaw the use of Rudd as a live bait. As a result, its rapid spread appears to have slowed.

Although there is no evidence that Rudd has been introduced to California, Dill and Cordone (1997) expressed concern that Rudd may find its way into the state as a contaminant in Golden Shiner shipments imported as bait from Arkansas. This species has been cultured in Arkansas and Virginia (and possibly elsewhere) as baitfish and distributed to bait stores in at least 16 states (Courtney and Williams 1992). Two specimens were taken from the Canadian side of Lake Erie at Crystal Beach (near Port Abino) in June 1997 (A. Dextrose, personal communication). That record is the first report of Rudd from Lake Erie.

Guinan et al. (2015) reported a similar summer trophic position to introduced Common Carp (*Cyprinus carpio*) and Goldfish (*Carassius auratus*), and suggested that seasonal trophic flexibility as a potential driver to facilitate both introduction success and the creation of novel nutrient transfer pathways among habitat types.

Voucher specimens: Massachusetts (MCZ 95616, MCZ 96072, MCZ 99432), New York (AMNH 58445, NYSM 25122, 35122, 40871, 41752, 41850), Vermont (NYSM 42040).

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US Fish and Wildlife Service Ecological Risk Screening Summary for *Scardinius erythrophthalmus*

[FishBase Summary](#)

Revision Date: 8/3/2021

Peer Review Date: 8/3/2021

Nixon, L. P., Miller, G., Jacobs, J., Larson, T., Haddad, A., Usato, M., Neilson, and A. Barros, 2025. *Scardinius erythrophthalmus* (Linnaeus, 1758): U.S. Geological Survey, Nonindigenous Aquatic Species Database, Gainesville, FL, <https://nas.er.usgs.gov/queries/FactSheet.aspx?SpeciesID=648>, Revision Date: 8/3/2021, Peer Review Date: 8/3/2021, Access Date: 12/8/2025

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